

MARK LAWRENCE

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PROFESSIONAL SUMMARY

Seasoned Software Engineering Technical Leader with over three decades of experience specializing in designing, evolving, and scaling complex, distributed full stack enterprise systems using C# and the .NET ecosystem. Proven expertise in defining and leading architecture across the full Software Development Lifecycle (SDLC). Adept at providing technical leadership in cross-functional teams, and mentoring talent within high-performing teams.

CORE COMPETENCIES & TECHNICAL SKILLS

Frontend: React.js, Angular, Blazor WASM, HTML, CSS, JavaScript, TypeScript

Backend & Frameworks: C#, .NET 5+, .NET Core, ASP.NET Core, Entity Framework Core, ASP.NET MVC, .Net Framework, Python, OAuth2 & OpenID Connect

Data & Persistence: Entity Framework Core, SQL Server, Cosmos DB, SQL, and ER Data Modelling

Architecture & Design: Microservices, Domain Driven Design (DDD), Clean Architecture, CQRS, Event Sourcing, SOA, Enterprise Application Architecture, System Migration

Cloud & DevOps: Microsoft Azure (App Services, Azure Functions, Azure Kubernetes Service (AKS), Azure SQL), Azure DevOps, GitHub, CI/CD, Team Foundation Server (TFS), MS Build, Octopus Deploy

PROFESSIONAL EXPERIENCE

Lead Software Engineer | Tobii Dynavox, Redondo Beach, CA (Remote) | 07/2022 – 08/2025

- **Technical Leadership & Strategy:** Provided strategic technical leadership to a team of 13 engineers in developing the Boardmaker software product, a critical special education solution serving over 900,000 users.
- **Architectural Modernization:** Initiated and led the future-state migration of desktop applications from a legacy Electron-based design to a modern, future-proof architecture based on .Net MAUI.
- **Process & Quality Improvement:** Led the engineering team in establishing and refining modern development practices, including implementing Git flow and semantic versioning automated via CI/CD pipelines to ensure superior code quality and version control.
- **AI Assisted Coding Tooling:** Participated in Tobii Dynavox pilot adoption program for GitHub CoPilot, later introducing the tool to the Boardmaker team, and establishing usage best practices.
- **Cloud & Cost Optimization:** Managed and optimized MS Azure resource allocation, driving significant cost-saving efforts, including a reduction of over \$8,000 per month in database costs and projecting savings exceeding \$65,000 annually for web image search services.
- **DevOps & Security Posture:** Overhauled the Azure DevOps build and deployment pipelines, integrating GitHub Advanced Security scans to enhance deployment reliability and significantly raise the team's security posture.
- **Risk Management:** Took ownership of security remediation efforts, successfully implementing updates to resolve more than 30 high-priority security concerns identified in a third-party audit, mitigating critical risks.
- **Cross-Functional Coordination:** Acted as the primary technical liaison between engineering teams, effectively coordinating complex work and ensuring alignment across all project phases.

Development Lead | ESL (Eyede Solutions Limited), Lower Hutt, New Zealand | 01/2013 – 07/2022

- **Microservices Architecture:** Architected and led the development of a next-generation web application, replacing a legacy ASP.Net MVC monolith with a Microservices backend based on .Net 6 and Blazor Web Assembly for the frontend.
- **Pattern Implementation:** Implemented advanced patterns including CQRS and Event Sourcing for the backend services, ensuring high decoupling and temporal consistency.
- **Cloud Native Deployment:** Deployed the system on Azure, utilizing Azure Kubernetes Services (AKS) and Azure App Services for orchestration, and RabbitMQ for reliable intra-service messaging.
- **Security Implementation:** Implemented and administered an Identity Server instance, providing robust OAuth2 and OpenID Connect based authentication and authorization for public-facing applications.
- **Business Impact:** Led the development of a trip processing service that resulted in a 600% increase in throughput compared to the legacy product.

- **DevOps Automation:** Configured and managed internal Team Foundation Server (TFS) deployment before migrating this capability to Azure DevOps, developing the standardized DevOps processes and procedures.
- **Cross-Customer Delivery:** Led the development of next-generation multi-tier web applications for two other customers, using ASP.Net Core, Entity Framework Core, and SQL Server, successfully replacing legacy solutions.

Solutions Architect | Optimation, Wellington, New Zealand | 05/2007 – 01/2013

- **Architectural Leadership:** Served as the Technical Lead and Architect for numerous .NET projects, defining and enforcing the archetype reference architecture across the organization.
- **Enterprise Backend Development:** Built and implemented reusable .NET software architecture and reference implementations using C#, ASP .Net MVC, and SQL Server with Entity Framework for data persistence.
- **Advanced Design Patterns:** Applied complex architectural principles, including Domain Driven Design (DDD) and Command Query Responsibility Segregation (CQRS), to ensure high-performance, scalable system design.
- **Full Stack Implementation:** Designed and delivered complete solutions, including front-end web pages using ASP .Net MVC, HTML5, JavaScript, jQuery, and CSS3 with Bootstrap.
- **Strategic Consulting:** Performed high-level technical architecture and delivery reviews for high-profile clients, including New Zealand's state-owned bank, Kiwibank, successfully securing additional downstream work for Optimation resources.
- **Process & Governance:** Configured, administered, and managed the internal Team Foundation Server environment, including process customization and end-user training, establishing foundational governance for development teams.
- **Pre-Sales Strategy:** Designed technical solutions and established accurate effort estimates during pre-sales engagements, driving successful client acquisition

Application Architect | EDS (New Zealand) Ltd, Wellington, New Zealand | 04/1998 – 05/2007

- **Enterprise Application Strategy:** Served as the Application Architect for a major New Zealand government client, responsible for end-to-end design and overall strategy of large-scale application systems.
- **Global Leadership:** Served as technical lead for EDS' New Zealand .Net Center of Expertise, and with other technical leaders on EDS' global .Net practice leadership team.
- **Standards & Governance:** Composed and authored the global C# coding standards document for the .NET practice and acted as the reviewer for the global VB.Net coding standards, ensuring code consistency across the region.
- **Process & Tools:** Served as the Team Foundation Server (TFS) expert, managing general administration, process customization, and end-user training for the entire team.
- **Talent Development:** Established and led a graduate recruitment program for the .NET practice, actively mentoring and interviewing successful candidates.

Senior Software Developer | Queensland Rail, Brisbane, Australia | 05/1997 – 02/1998

- Developed and enhanced client/server software projects utilizing Visual Basic and 4GL programming languages.
- Provided second-level support to end users of several key software tools.

Analyst/Programmer | EDS (New Zealand) Ltd, Wellington, New Zealand | 06/1994 – 02/1997

- **Mainframe Programming:** Developed and enhanced core functionality using CICS COBOL and Assembler language, directly supporting major banking clients, following completion of an in-house training program.
- **On-Call Support:** Supported critical operations for one of EDS' major banking clients as part of the on-call team

EDUCATION

Bachelor of Science | Victoria University of Wellington, Wellington, New Zealand

CS50P Certificate Python Course | Harvard University (Online)